

प्रा. मोटेगावकर सरांचे

RCC

एकाग्रता हा सर्व प्रकारच्या ज्ञानाचा पाया आहे.
सामान्य माणूस नव्वद टक्के विचारशक्ती वाया
घालवतो,



म्हणूनच त्याच्या हातून घोडयुका होतात.
ज्याच्या मनाला वळण लागलेलं आहे
असा मनुष्य कधीही चूक करणार नाही.

-स्वामी विवेकानंद

Mark
720

Group
PCB

Repeater NEET (2025-26)

PCB Test

Date : 17/08/2025

Time : 3.00 Hrs

Physics -45

Chemistry -45

Biology -90

Question Booklet Version

11

(Write this number on
your Answer Sheet)

Roll Number

Question Booklet Sr. No.

• Today's Test Syllabus •

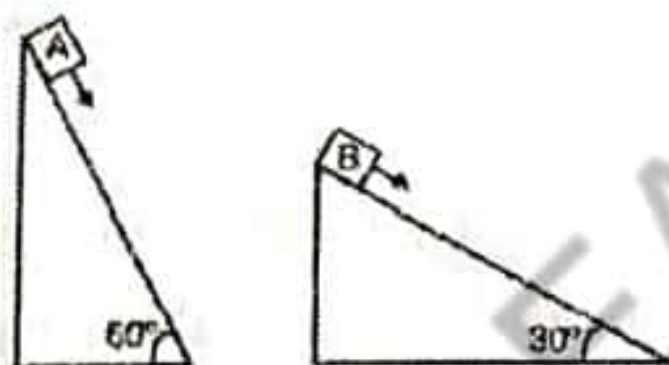
Physics : 1D + 2D + Thermodynamics + Dual Nature

Chemistry : Isomerism + Peri. Table

Biology : Excretory Products + Locomotion + Respiration + Photosynthesis

SECTION 'A' PHYSICS

01. Two fixed frictionless inclined planes making an angle 30° and 60° with the vertical are shown in the figure. Two blocks A and B are placed on the two planes. What is the relative vertical acceleration of A with respect to B?



- 1) 4.9 ms^{-2} in horizontal direction
2) 9.8 ms^{-2} in vertical direction
3) Zero
4) 49 ms^{-2} in vertical direction
02. Three elephants A, B and C are moving along a straight line with constant speed in same direction as shown in figure. Speed of A is 5 m/s and speed of C is 10 m/s. Initially separation between A & B is 'd' and between B & C is also d. When 'B' catches 'C' separation between A & C becomes 3d. Then the speed of B will be -



- 1) 7.5 m/s
2) 15 m/s
3) 20 m/s
4) 5 m/s

03. A man in a balloon, throws a stone downwards with a speed of 5 m/s with respect to balloon. The balloon is moving upwards with a constant acceleration of 5 m/s^2 . Then velocity of the stone relative to the man after 2 second is



- 1) 10 m/s
2) 30 m/s
3) 15 m/s
4) 35 m/s

PCB TEST

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04. A coin is released inside a lift at a height of 2 m from the floor of the lift. The height of the lift is 10 m. The lift is moving with an acceleration of 11 m/s^2 downwards. The time after which the coin will strike with the floor of lift is
- 4s
 - 2s
 - $\frac{4}{\sqrt{21}} \text{ s}$
 - Never
05. A ball A is dropped from a building of height 45 m. Simultaneously another identical ball B is thrown up with a speed 50 m/s . The relative speed of ball B w.r.t. ball A at any instant of time is (Take $g = 10 \text{ m/s}^2$)
- 0
 - 10 m/s
 - 25 m/s
 - 50 m/s
06. A ball A is thrown vertically upwards with speed u . At the same instant another ball B is released from rest at height h . At time t , the speed of A relative to
- u
 - $u - 2gt$
 - $\sqrt{u^2 - 2gh}$
 - $u - gt$
07. Two cars A and B are running at velocities of 60 km/h and 45 km/h . What is the relative velocity of car A with respect to car B, if both are moving eastward?
- 15 km/h
 - 45 km/h
 - 60 km/h
 - 105 km/h

08. On a two-lane road, Car A is travelling with a speed of 36 km/h . Two cars B and C approach car A in opposite directions with a speed of 54 km/h each. At a certain instant, when the distance AB is equal to AC, both being 1 km, B decides to overtake A before C does. The minimum required acceleration of car B to avoid an accident is
- 1 m/s^2
 - 1.5 m/s^2
 - 2 m/s^2
 - 3 m/s^2

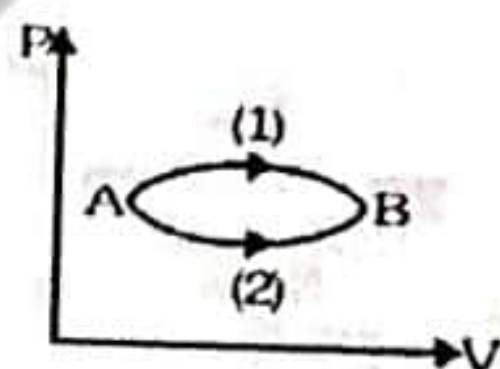
09. Statement-1 : A gas is taken from state A to state B through two different paths.

Molar specific heat capacity in path (1) is more as compared to (2). and

Statement-2 : Specific heat

$$C = \frac{Q}{n\Delta T} \text{ \& } Q = \Delta U + W \text{ and } W \text{ is equal to}$$

area under P-V diagram.



- Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
- Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1
- Statement-1 is True, Statement-2 is False.
- Statement-1 is False, Statement-2 is True.

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10. A bird is tossing (flying to and fro) between two cars moving towards each other on a straight road. One car has speed of 27 km h^{-1} while the other has the speed of 18 km h^{-1} bird starts moving from first car towards the Other and is moving with the speed of 36 km h^{-1} when the two cars were separated by 36 km . The total distance covered by the bird is
- 28.8 km
 - 38.8 km
 - 48.8 km
 - 58.8 km
11. Light of two different frequencies whose photons have energies of 1 eV and 2.5 eV are incident one by one on a metal surface of work function 0.5 eV . The ratio of maximum energies of emitted electrons will be
- $1 : 4$
 - $4 : 1$
 - $1 : 2$
 - $2 : 1$
12. A 175 m long train is traveling along a straight track with a velocity 72 km h^{-1} . A bird is flying parallel to the train in the opposite direction with a velocity 18 km h^{-1} . The time taken by the bird to Cross the train is
- 35 s
 - 27 s
 - 11.6 s
 - 7 s
13. Two parallel rail tracks run north-south. On onetrack train A moves north with a speed of 54 km h^{-1} and on the other track train B moves south with a speed of 90 km h^{-1} The velocity of train A respect to train B is
- 10 m s^{-1}
 - 15 m s^{-1}
 - 25 m s^{-1}
 - 40 m s^{-1}
14. A jet airplane travelling at the speed of 500 km h^{-1} ejects its products of combustion at the speed of 1500 km h^{-1} . relative to the jet plane. The speed of the products of combustion with respect to an observer on the ground is
- 500 km h^{-1}
 - 1000 km h^{-1}
 - 1500 km h^{-1}
 - 2000 km h^{-1}
15. Which one of the following represents displacement-time graph of two objects A and B moving with zero relative velocity?
- -
 -
 -
16. At a metro station. a girl walks up a stationary escalator in time t_1 . If she remains stationary on the escalator, then the escalator take her up in time t_2 . The time taken by her to walk up on the moving escalator will be
- $\frac{(t_1 + t_2)}{2}$
 - $\frac{t_1 t_2}{(t_2 - t_1)}$
 - $\frac{t_1 t_2}{(t_2 + t_1)}$
 - $t_1 - t_2$

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17. Suppose that two objects A and B are moving with velocities $\vec{v}_A$ and $\vec{v}_B$ (each with respect to some common frame of reference). Let $\vec{v}_{AB}$ represent the velocity of A with respect to B. Then
- 1) $\vec{v}_{AB} + \vec{v}_{BA} = 0$
 - 2) $\vec{v}_{AB} - \vec{v}_{BA} = 0$
 - 3) $\vec{v}_{AB} = \vec{v}_A + \vec{v}_B$
 - 4) $|\vec{v}_{AB}| \neq |\vec{v}_{BA}|$
18. Photo electric effect is the phenomenon in which
- 1) Photons come out of a metal when it is hit by a beam of electrons
 - 2) Photons come out of the nucleus of an atom under the action of an electric field
 - 3) Electrons come out of metal with a constant velocity depending on frequency and intensity of incident light
 - 4) Electrons come out of a metal with different velocity not greater than a certain value which depends only on the frequency of the incident light wave and not on its intensity.
19. When an electro-magnetic radiation is incident on the surface of metal, maximum kinetic energy of photoelectron depends on-
- 1) Frequency of radiation
 - 2) Intensity of radiation
 - 3) Both the frequency and intensity
 - 4) Polarization of radiation
20. The work functions of tungsten and sodium are 5.06 eV and 2.53 eV respectively. If the threshold wavelength for sodium is 5896 Å, then the threshold wavelength for the tungsten will be
- 1) 11792 Å
 - 2) 5896 Å
 - 3) 4312 Å
 - 4) 2948 Å
21. The stopping potential necessary to reduce the photoelectric current to zero-
- 1) is directly proportional to wavelength of incident light.
 - 2) uniformly increases with the wavelength of incident light.
 - 3) directly proportional to frequency of incident light.
 - 4) uniformly increases with the frequency of incident light.
22. Photoelectric effect takes place in element A. Its work function is 2.5 eV and threshold wavelength is λ . An other element B is having work function of 5 eV. Then find out the wavelength that can produce photoelectric effect in B.
- 1) $\lambda/2$
 - 2) 2λ
 - 3) λ
 - 4) 3λ
23. When a certain metallic surface is illuminated with monochromatic light of wavelength λ , the stopping potential for photo electric current is $6 V_0$. When the same surface is illuminated with light of wavelength 2λ , the stopping potential is $2V_0$. The threshold wavelength of this surface for photoelectric effect is
- 1) 6λ
 - 2) $4\lambda/3$
 - 3) 4λ
 - 4) 8λ
24. A photo sensitive metallic surface has work function $h\nu_0$. If photons of energy $2h\nu_0$ fall on this surface, the electrons come out with a maximum velocity of 4×10^6 m/s. When the photon energy is increased to $5 h\nu_0$, then maximum velocity of photo electrons will be-
- 1) 2×10^6 m/s
 - 2) 2×10^7 m/s
 - 3) 8×10^5 m/s
 - 4) 8×10^6 m/s

Space For Rough Work

25. Statement-1 : Though light of a single frequency (monochromatic) is incident on a metal, the energies of emitted photoelectrons are different. and

Statement-2 : The energy of electrons emitted from inside the metal surface, is lost in collision with the other atoms in the metal.

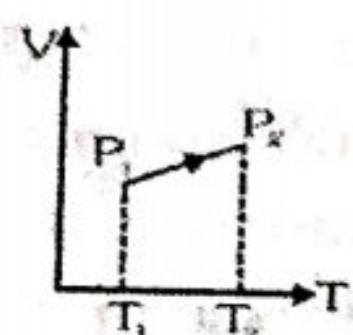
1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

2) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1

3) Statement-1 is True, Statement-2 is False.

4) Statement-1 is False, Statement-2 is True.

26. From the following V-T diagram we can conclude:-



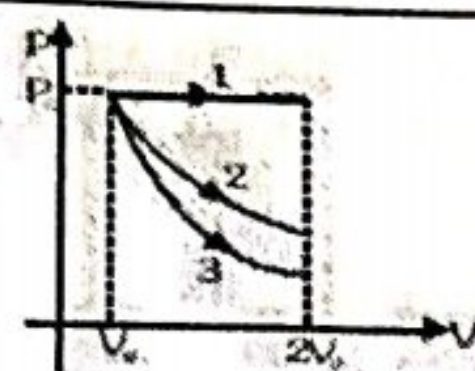
1) $P_1 = P_2$

2) $P_1 > P_2$

3) $P_1 < P_2$

4) Can't say anything

27. A gas is expanded from volume V_0 to $2V_0$ under three different processes. Process 1 is isobaric process, process 2 is isothermal and process 3 is adiabatic. Let ΔU_1 , ΔU_2 and ΔU_3 be the change in internal energy of the gas in these processes. Then :-



1) $\Delta U_1 > \Delta U_2 > \Delta U_3$

2) $\Delta U_1 < \Delta U_2 < \Delta U_3$

3) $\Delta U_2 < \Delta U_1 < \Delta U_3$

4) $\Delta U_2 < \Delta U_3 < \Delta U_1$

28. For an ideal gas $PT^n = \text{constant}$ then volume expansion coefficient is equal to :-

1) $\frac{11}{T}$

2) $\frac{1}{T}$

3) $\frac{12}{T}$

4) $\frac{2}{T}$

29. The internal energy of a gas is given by

$U = 5 + 2PV$. It expands from V_0 to $2V_0$ against a constant pressure P_0 . The heat absorbed by the gas in the process is :-

1) $-3P_0V_0$

2) $3P_0V_0$

3) $2P_0V_0$

4) P_0V_0

30. When water is heated from 0°C to 4°C and C_p and C_v are its specific heats at constant pressure and constant volume respectively, then :-

1) $C_p > C_v$

2) $C_p < C_v$

3) $C_p = C_v$

4) $C_p - C_v = R$

31. The molar specific heat of the process $V \propto T^4$ for CH_4 gas at room temperature is:-

1) $4R$

2) $7R$

3) $3R$

4) $8R$

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32. The internal energy of a gas in an adiabatic process is given by $U = a + bPV$, find γ :-

1) $\frac{a+1}{a}$

2) $\frac{b+1}{a}$

3) $\frac{b+1}{a}$

4) $\frac{a}{b+1}$

33. Statement-1 : On sudden expansion a gas cools. and

Statement-2 : On sudden expansion, no heat is supplied to system and hence gas does work at the expense of its internal energy.

1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

2) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1

3) Statement-1 is True, Statement-2 is False.

4) Statement-1 is False, Statement-2 is True.

34. Rain is falling vertically with a speed of 30 m s^{-1} . A woman rides a bicycle with a speed of 12 m s^{-1} in east to west direction. In direction she should hold her umbrella?

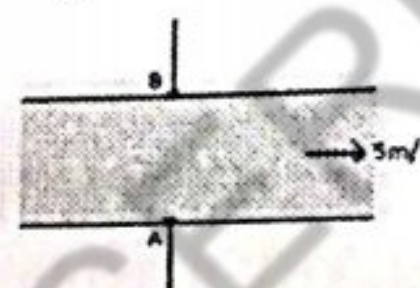
1) At an angle of $\tan^{-1}\left(\frac{2}{5}\right)$ with the vertical towards the east.

2) At angle of $\tan^{-1}\left(\frac{2}{5}\right)$ with the vertical towards the west.

3) At angle of $\tan^{-1}\left(\frac{5}{2}\right)$ with the vertical towards the east.

4) At angle of $\tan^{-1}\left(\frac{5}{2}\right)$ with the vertical towards the west.

35. A river is flowing from west to east with a speed 5 m s^{-1} . A swimmer can swim in still water at a speed of 10 m s^{-1} . If he wants to start from point A on south bank and reach opposite point B on north he swim?



1) 30° east of north

2) 60° east of north

3) 30° west of north

4) 60° west of north

36. A girl riding a bicycle with a speed of 5 m s^{-1} towards north direction, observes rain falling vertically down. If she increases her speed to 10 m s^{-1} , rain appears to meet her at 45° to the vertical. What is the speed of the rain?

1) $5\sqrt{2} \text{ m s}^{-1}$

2) 5 m s^{-1}

3) $10\sqrt{2} \text{ m s}^{-1}$

4) 10 m s^{-1}

37. The motion of a body is given by the equation $\frac{dv}{dt} = 6 - 3v$ where v is the speed in m s^{-1} and t is time in s. The body is at rest at $t=0$. The speed varies with time as

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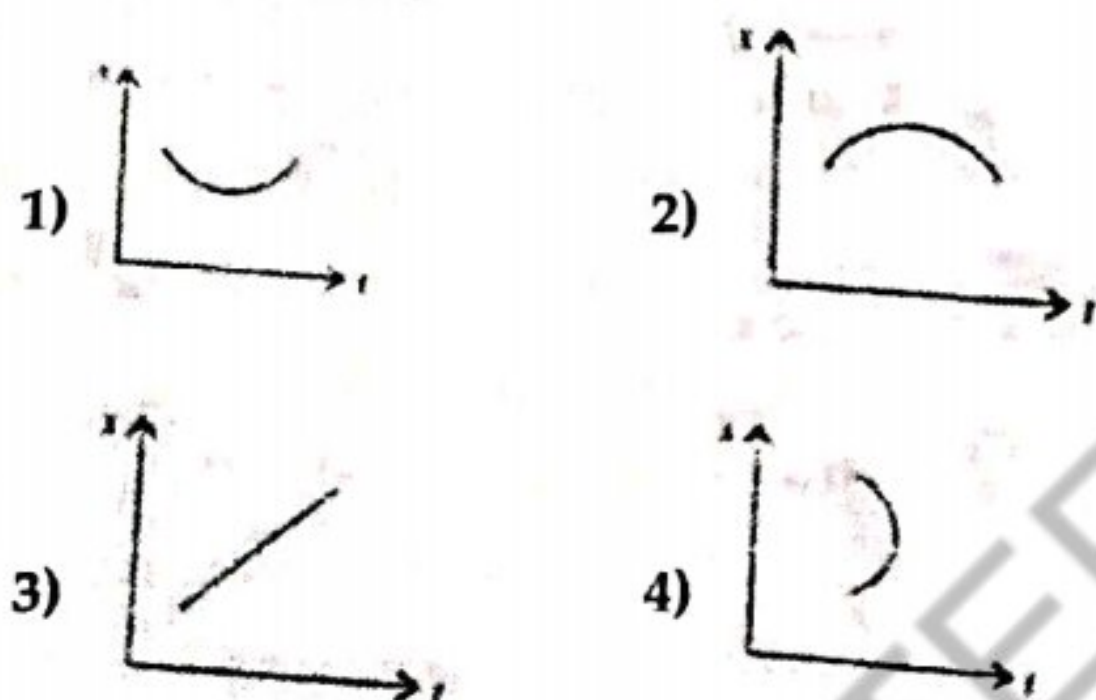
- 1) $v=(1-e^{-3t})$ 2) $v=2(1-e^{-3t})$
 3) $v=(1+e^{-2t})$ 4) $v=2(1+e^{-2t})$

38. Statement-1 : In a photoelectric effect experiment different photoelectrons have different kinetic energy. and

Statement-2 : In the incident radiation different photons must have different energy.

- 1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
 2) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1
 3) Statement-1 is True, Statement-2 is False.
 4) Statement-1 is False, Statement-2 is True.

39. Position-time graph for motion with zero acceleration is



40. Which of the following statements is not correct?

- 1) The zero velocity of a body at any instant does not necessarily imply zero acceleration at that instant.
 2) The kinematic equation of motions are true

only for motion in which the magnitude and the direction of acceleration are constants during the course of motion.

- 3) The sign of acceleration tells us whether the particle's speed is increasing or decreasing.
 4) All of these

41. A particle if, released from rest from a tower of height $3h$. The ratio of the intervals of time to cover three equal heights h is

- 1) $t_1 : t_2 : t_3 = 3 : 2 : 1$
 2) $t_1 : t_2 : t_3 = 1 : (\sqrt{2}-1) : (\sqrt{3}-2)$
 3) $t_1 : t_2 : t_3 = \sqrt{3} : \sqrt{2} : 1$
 4) $t_1 : t_2 : t_3 = 1 : (\sqrt{2}-1) : (\sqrt{3}-\sqrt{2})$

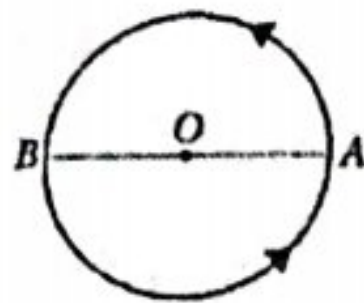
42. A motorcycle and a car start from rest from the same place at the same time and travel in the same direction. The motorcycle accelerates at 1.0 m s^{-2} up to a speed of 36 km h^{-1} and the car at 0.5 m s^{-2} up to a speed $d \text{ km h}^{-1}$ time at which the car would overtake the motorcycle is

- 1) 20s 2) 23 s
 3) 30s 4) 35 s

43. A particle moving on a circular path of radius R constant with speed v . During motion of the particle from point A to point B

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- 1) Average speed is $v/2$
 - 2) The magnitude of average velocity is $\frac{v}{\pi}$
 - 3) The magnitude of average acceleration is $\frac{2v^2}{\pi R}$
 - 4) Average velocity is zero
44. Statement-1 : The isothermal curves intersect each other at a certain point. and
Statement-2 : The isothermal change are done slowly, so the isothermal curves have very little slope.
- 1) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
 - 2) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1
 - 3) Statement-1 is True, Statement-2 is False.
 - 4) Statement-1 is False, Statement-2 is True.
45. A train is standing on a platform, a man inside a compartment of a train drops a stone. At the same instant train starts to move with constant acceleration The path of the particle as seen by the person who drops the stone is
- 1) parabola
 - 2) straight line for sometime & parabola for the remaining time

- 3) straight line
- 4) variable path that cannot be defined.

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SECTION 'B' CHEMISTRY

46. Match List-I with List-II :

List-I (Elements)	List-II (Properties in their respective groups)
A) Cl, S	I) Elements with highest electronegativity
B) Ge, As	II) Elements with largest atomic size
C) Fr, Ra	III) Elements which show properties of both metals and non-metal
D) F, O	IV) Elements with highest negative electron gain enthalpy

Choose the correct answer from the options given below :

- 1) A-(II), B-(III), C-(IV), D-(I)
- 2) A-(III), B-(II), C-(I), D-(IV)
- 3) A-(IV), B-(III), C-(II), D-(I)
- 4) A-(II), B-(I), C-(IV), D-(III)

47. In comparison to boron, beryllium has :

- 1) Lesser nuclear charge and lesser first ionisation enthalpy
- 2) Greater nuclear charge and lesser first ionisation enthalpy
- 3) Greater nuclear charge and greater first ionisation enthalpy
- 4) Lesser nuclear charge and greater first ionisation enthalpy

48. Match List-I with List-II :

List-I	List-II
A) $Al^{3+} < Mg^{2+} < Na^+ < F^-$	I) Ionisation enthalpy
B) $B < C < O < N$	II) Metallic character
C) $B < Al < Mg < K$	III) Electronegativity
D) $Si < P < S < Cl$	IV) Ionic radii

Choose the correct answer from the options given below :

- 1) A-(IV), B-(I), C-(III), D-(II)
- 2) A-(II), B-(III), C-(IV), D-(I)
- 3) A-(IV), B-(I), C-(II), D-(III)
- 4) A-(III), B-(IV), C-(II), D-(I)

49. The successive 5 ionisation energies of an element are 800, 2427, 3658, 25024 and 32824 kJ/mol, respectively. By using the above values predict the group in which the above element is present :

- 1) Group 2
- 2) Group 13
- 3) Group 4
- 4) Group 14

50. Which of the following statements are correct?

- A) The process of the addition an electron to a neutral gaseous atom is always exothermic
- B) The process of removing an electron from an isolated gaseous atom is always endothermic
- C) The 1st ionization energy of the boron is less than that of the beryllium
- D) The electronegativity of C is 2.5 in CH_4 and CCl_4
- E) Li is the most electropositive among elements of group I

- 1) B and C only
- 2) A, C and D only
- 3) B and D only
- 4) B, C and E only

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51. Match List I with List II

List-I		List-II	
A)	37	I)	p-block
B)	78	II)	d-block
C)	52	III)	f-block
D)	65	IV)	s-block

Choose the correct answer from the options given below :

- 1) A-(II), B-(IV), C-(I), D-(III)
- 2) A-(I), B-(III), C-(IV), D-(II)
- 3) A-(IV), B-(III), C-(II), D-(I)
- 4) A-(IV), B-(II), C-(I), D-(III)

52. The correct increasing order of the ionic radii is

- 1) $\text{Cl}^- < \text{Ca}^{2+} < \text{K}^+ < \text{S}^{2-}$
- 2) $\text{K}^+ < \text{S}^{2-} < \text{Ca}^{2+} < \text{Cl}^-$
- 3) $\text{S}^{2-} < \text{Cl}^- < \text{Ca}^{2+} < \text{K}^+$
- 4) $\text{Ca}^{2+} < \text{K}^+ < \text{Cl}^- < \text{S}^{2-}$

53. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion (A) : The first ionisation enthalpy decreases across a period.

Reason (R) : The increasing nuclear charge outweighs the shielding across the period.

In the light of the above statements, choose the most appropriate from the options given below :

- 1) A is false but R is true
- 2) A is true but R is false
- 3) Both A and R are true and R is the correct explanation of A
- 4) Both A and R are true but R is NOT the correct explanation of A

54. Given below are two statements

Statement I : The electronegativity of group 14 elements from Si to Pb, gradually decreases.

Statement II : Group 14 contains non-metallic, metallic, as well as metalloid elements.

In the light of the above statements, choose the most appropriate from the options given below :

- 1) Statement I is true but Statement II is false
- 2) Statement I is false but Statement II is true
- 3) Both Statement I and Statement II are false
- 4) Both Statement I and Statement II are true

55. Given below are two statements :

Statement I : The correct order of first ionization enthalpy values of Li, Na, F and Cl is $\text{Na} < \text{Li} < \text{Cl} < \text{F}$.

Statement II : The correct order of negative electron gain enthalpy values of Li, Na, F and Cl is $\text{Na} < \text{Li} < \text{F} < \text{Cl}$

In the light of the above statements, choose the correct answer from the options given below :

- 1) Both Statement I and Statement II are true
- 2) Both Statement I and Statement II are false
- 3) Statement I is false but Statement II is true
- 4) Statement I is true but Statement II is false

56. Which one of the following sets of ions represents a collection of isoelectronic species?

(Given : Atomic number : F : 9, Cl : 17, Na = 11, Mg = 12, Al = 13, K = 19, Ca = 20, Sc = 21)

- 1) $\text{N}^{3-}, \text{O}^{2-}, \text{F}^-, \text{S}^{2-}$
- 2) $\text{Ba}^{2+}, \text{Sr}^{2+}, \text{K}^+, \text{Ca}^{2+}$
- 3) $\text{K}^+, \text{Cl}^-, \text{Ca}^{2+}, \text{Sc}^{3+}$
- 4) $\text{Li}^+, \text{Na}^+, \text{Mg}^{2+}, \text{Ca}^{2+}$

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57. The incorrect statement is
- 1) The first ionization enthalpy of K is less than that of Na and Li
 - 2) Xe does not have the lowest first ionization enthalpy in its group
 - 3) The first ionization enthalpy of element with atomic number 37 is lower than that of the element with atomic number 38
 - 4) The first ionization enthalpy of Ga is higher than that of the d-block element with atomic number 30
58. Among Al_2O_3 , SiO_2 , P_2O_3 and SO_2 , the correct order of acid strength is
- 1) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{SO}_2 < \text{P}_2\text{O}_3$
 - 2) $\text{SiO}_2 < \text{SO}_2 < \text{Al}_2\text{O}_3 < \text{P}_2\text{O}_3$
 - 3) $\text{SO}_2 < \text{P}_2\text{O}_3 < \text{SiO}_2 < \text{Al}_2\text{O}_3$
 - 4) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_3 < \text{SO}_2$
59. For Na^+ , Mg^{2+} , F^- and O^{2-} , the correct order of increasing ionic radii is :
- 1) $\text{O}^{2-} < \text{F}^- < \text{Na}^+ < \text{Mg}^{2+}$
 - 2) $\text{Na}^+ < \text{Mg}^{2+} < \text{F}^- < \text{O}^{2-}$
 - 3) $\text{Mg}^{2+} < \text{Na}^+ < \text{F}^- < \text{O}^{2-}$
 - 4) $\text{Mg}^{2+} < \text{O}^{2-} < \text{Na}^+ < \text{F}^-$
60. Given below are two statements :
- Statement (I) : Both metals and non-metals exist in p and d-block elements.
- Statement (II) : Non-metals have higher ionisation enthalpy and higher electronegativity than the metals.
- In the light of the above statements, choose the most appropriate answer from the options given below :
- 1) Both Statement I and Statement II are false
 - 2) Both Statement I and Statement II are true
 - 3) Statement I is false but Statement II is true
 - 4) Statement I is true but Statement II is false
61. Given below are two statements about X-ray spectra of elements :
- Statement (I) : A plot of $\sqrt{\nu}$ (ν = frequency of X-rays emitted) vs atomic mass is a straight line.
- Statement (II) : A plot of ν (ν = frequency of X-rays emitted) vs atomic number is a straight line.
- In the light of the above statements choose the correct answer from the options given below :
- 1) Statement I is true but Statement II is false
 - 2) Both Statement I and Statement II are true
 - 3) Both Statement I and Statement II are false
 - 4) Statement I is false but Statement II is true
62. Group-13 elements react with O_2 in amorphous form to form oxides of type M_2O_3 (M = element). Which among the following is the most basic oxide ?
- 1) Al_2O_3
 - 2) Ga_2O_3
 - 3) Ti_2O_3
 - 4) B_2O_3
63. The acidic, basic and amphoteric oxides, respectively, are
- 1) Na_2O , SO_3 , Al_2O_3
 - 2) Cl_2O , CaO , P_4O_{10}
 - 3) N_2O_5 , Li_2O , Al_2O_3
 - 4) MgO , Cl_2O , Al_2O_3
64. Similarity in chemical properties of the atoms of elements in a group of the periodic table is most closely related to :
- 1) Atomic numbers
 - 2) Atomic masses
 - 3) Number of principal energy levels
 - 4) Number of valence electrons

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65. B has a smaller first ionization enthalpy than Be. Consider the following statements :

I) It is easier to remove 2p electron than 2s electron

II) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be

III) 2s electron has more penetration power than 2p electron

IV) atomic radius of B is more than Be (atomic number B = 5, Be = 4)

The correct statements are :

- 1) (I), (II) and (IV) 2) (II), (III) and (IV)
3) (I), (II) and (III) 4) (I), (III) and (IV)

66. The correct order of electron gain enthalpies of Cl, F, Te and Po is

- 1) $F < Cl < Te < Po$ 2) $Po < Te < F < Cl$
3) $Te < Po < Cl < F$ 4) $Cl < F < Te < Po$

67. Among the following basic oxide is

- 1) SO_3 2) SiO_2 3) CaO 4) Al_2O_3

68. Match List-I with List-II.

List-I (Oxide)	List-II (Nature)
A) Cl_2O_7	I) Amphoteric
B) Na_2O	II) Basic
C) Al_2O_3	III) Neutral
D) N_2O	IV) Acidic

Choose the correct answer from the options given below :

- 1) A-(IV), B-(III), C-(I) D-(II)
2) A-(IV), B-(II), C-(I), D-(III)
3) A-(II), B-(IV), C-(III), D-(I)
4) A-(I), B-(II), C-(III), D-(IV)

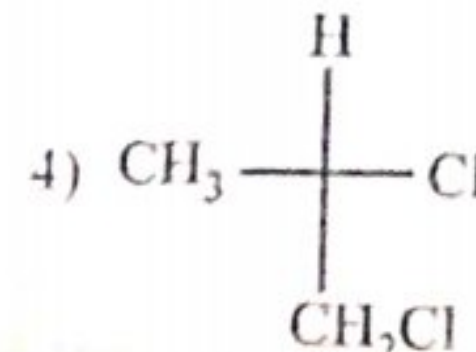
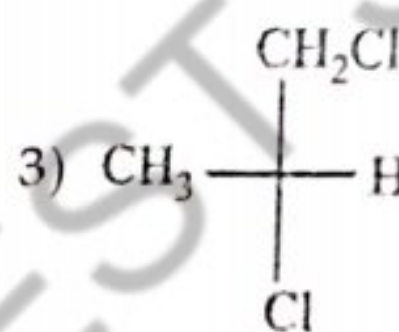
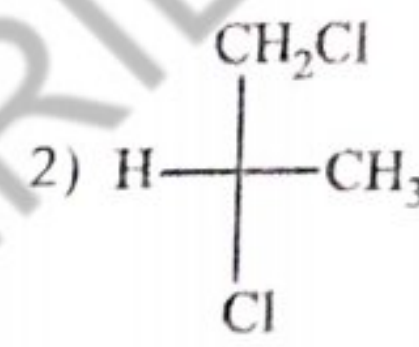
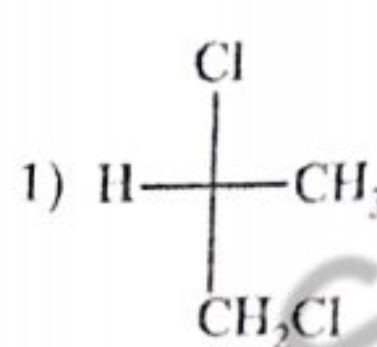
69. The ionic radii (in Å) of N^{3-} , O^{2-} and F^- are respectively

- 1) 1.71, 1.40 and 1.36 2) 1.71, 1.36 and 1.40
3) 1.36, 1.40 and 1.71 4) 1.36, 1.71 and 1.40

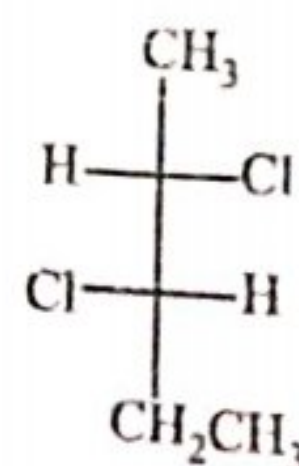
70. When the first electron gain enthalpy ($\Delta_{eg}H$) of oxygen is -141 kJ/mol , its second electron gain enthalpy is

- 1) A more negative value than the first
2) Almost the same as that of the first
3) Negative but less negative than the first
4) A positive value

71. Which of the following compounds has S configuration ?



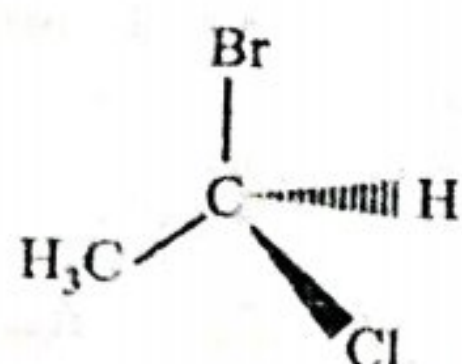
72. What is the configuration of the following molecule ?



- 1) 2S, 3R 2) 3R, 3S
3) 2S, 3S 4) 2R, 3R

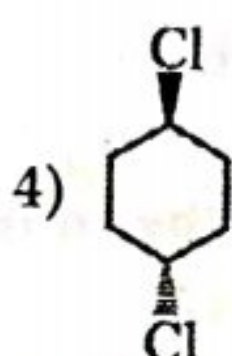
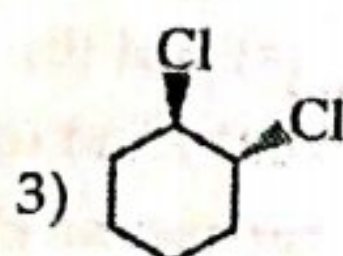
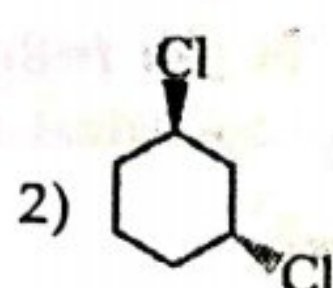
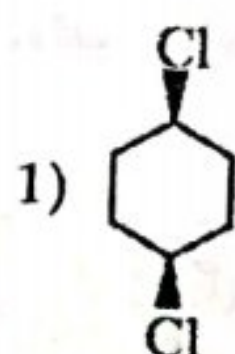
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73. The configuration of the given compound is

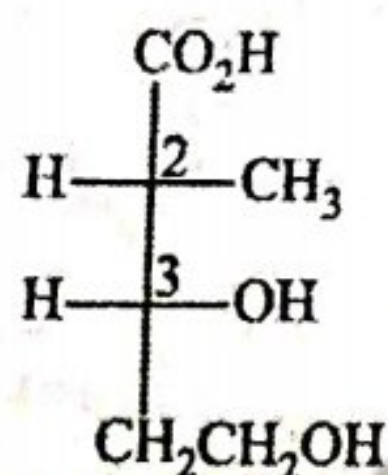


- 1) E
2) R
3) S
4) Z

74. Identify the compound in which COS is present

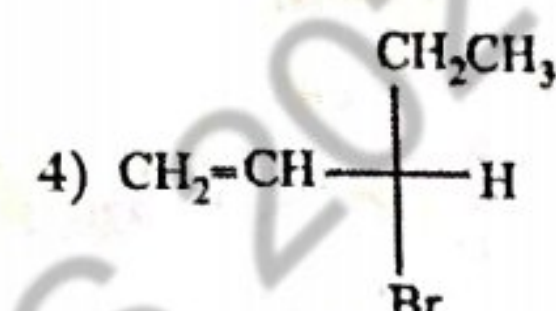
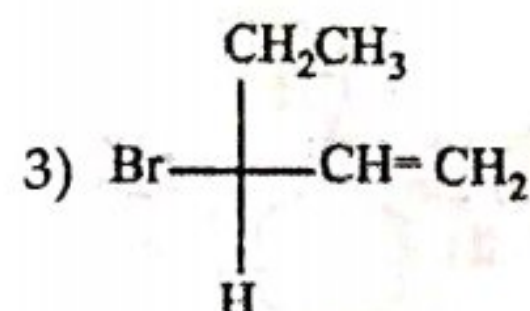
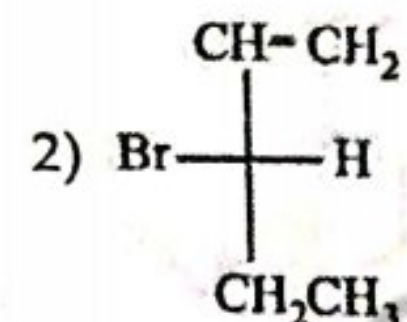
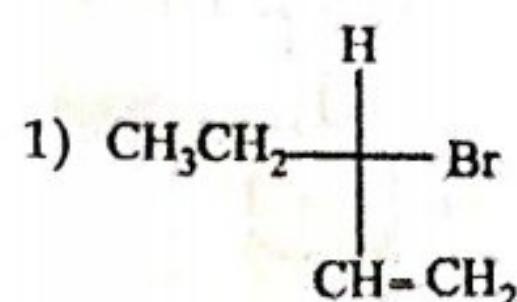


75. In the Fischer projection formula given below, what are the configurations of the two asymmetric centers ?

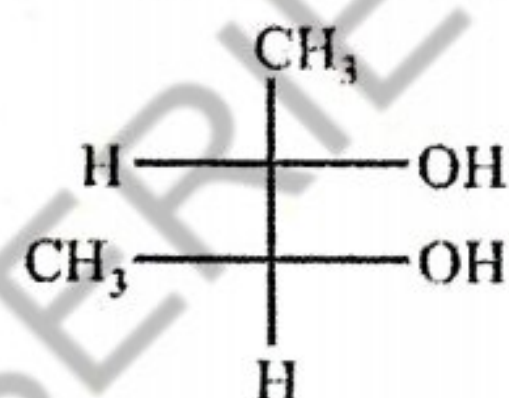


- 1) 2R, 3R
2) 2R, 3S
3) 2S, 3R
4) 2S, 3S

76. Which of the following compounds has R configuration ?

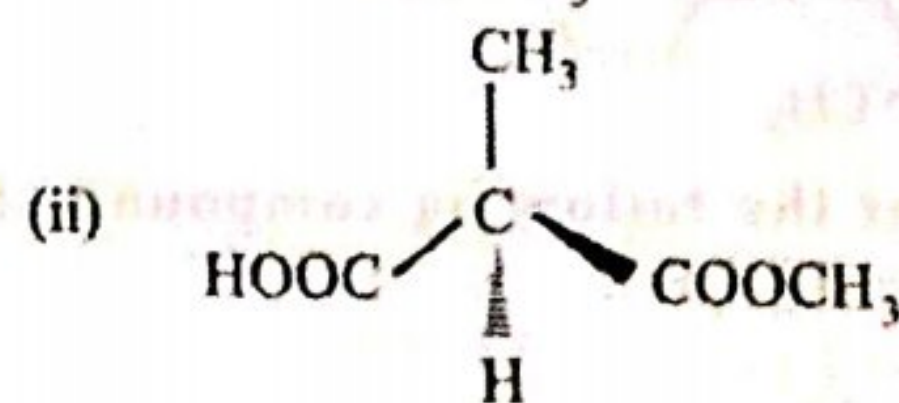
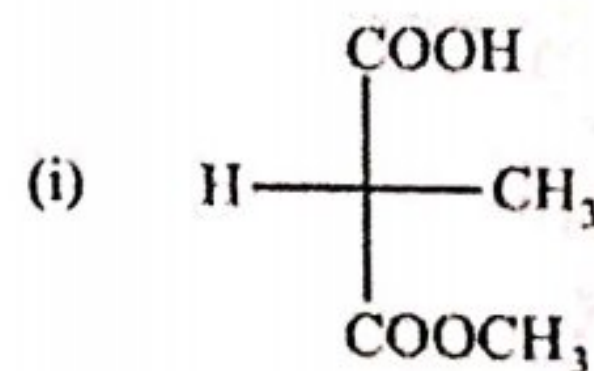


77. Correct configuration of the following is :



- 1) 2S, 3S
2) 2S, 3R
3) 2R, 3S
4) 2R, 3R

78. The correct configuration assigned for compound (i) and (ii) respectively is

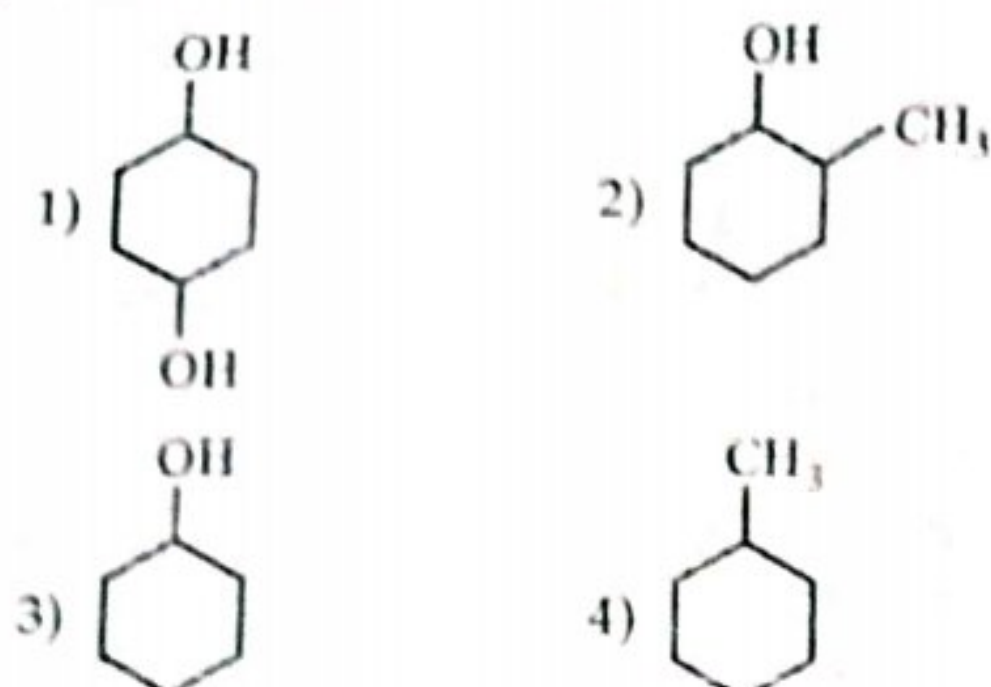


- 1) R, R
2) S, S
3) S, R
4) R, S

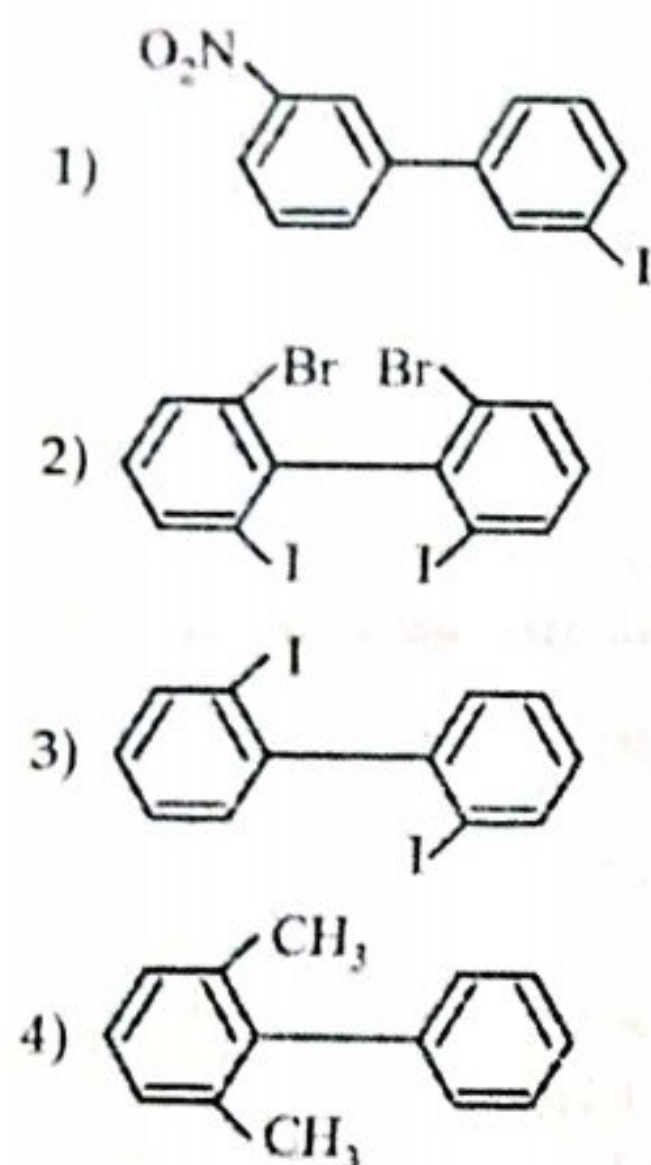
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79. Which of the following is optically active :



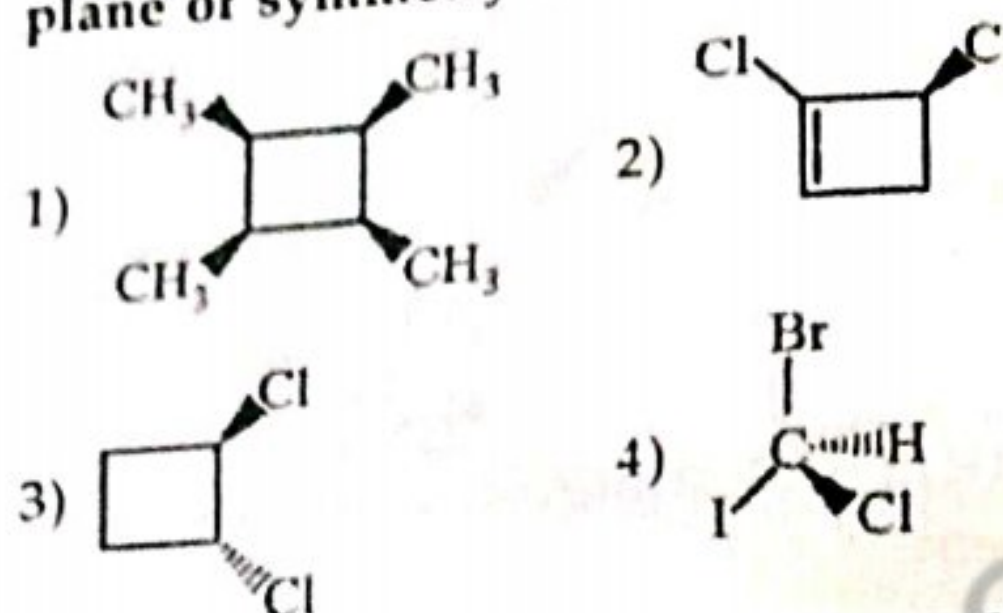
80. Which of the following biphenyls is optically active ?



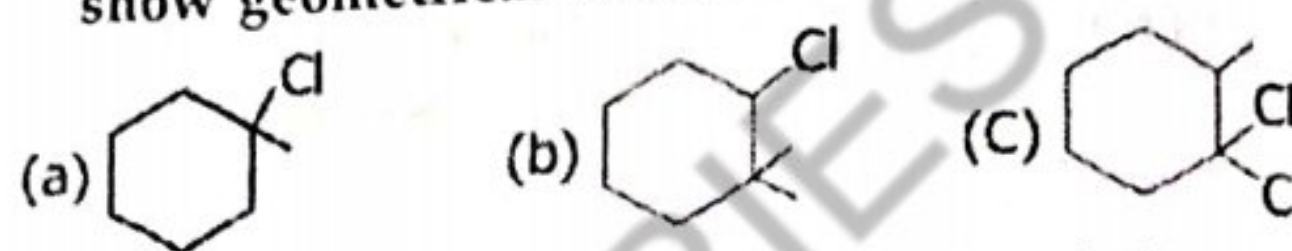
81. Which of the following compound shows optical isomerism ?

- 1) $\text{CH}_3\text{CH}_2\text{CH}_3$
- 2) $\text{CH}_2\text{OHCHOHCH}_2\text{OH}$
- 3) $\text{CH}_3\text{CHOHC}_2\text{H}_5$
- 4) CCl_2F_2

82. In which of the following compound have plane of symmetry ?



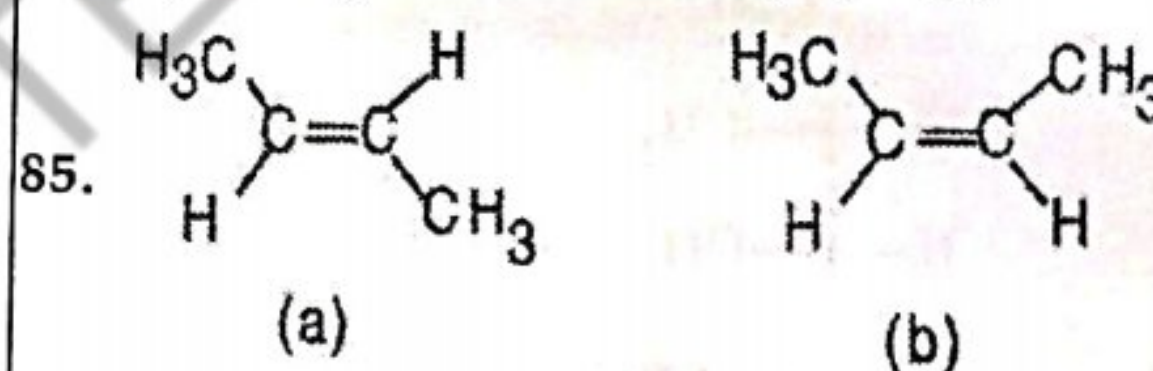
83. Which of the following compound will not show geometrical isomerism



- 1) Only (a)
- 2) Only (a) and (b)
- 3) Only (a) and (c)
- 4) All (a), (b) and (c)

84. Which of the following pair represent chain isomers

- 1) $\text{CH}_3\text{-CH}_2\text{-CHClBr}$ and $\text{ClCH}_2\text{-CH}_2\text{-CH}_2\text{Br}$
- 2) Isopentyl alcohol and neopentylalcohol
- 3) Pentan-1-al and 2-methyl butan-1-ol
- 4) Diethyl ether and Ethyl propyl ether



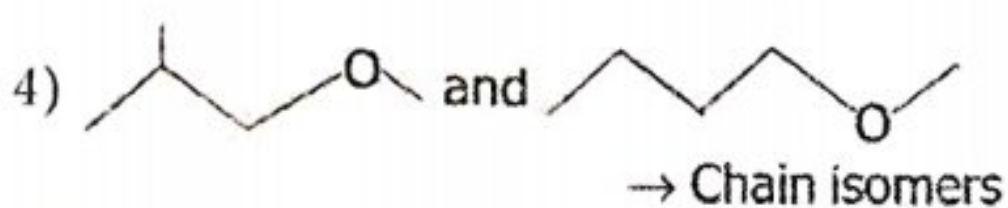
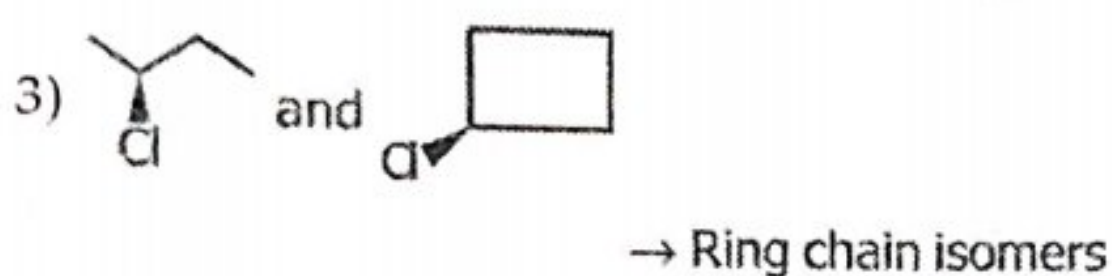
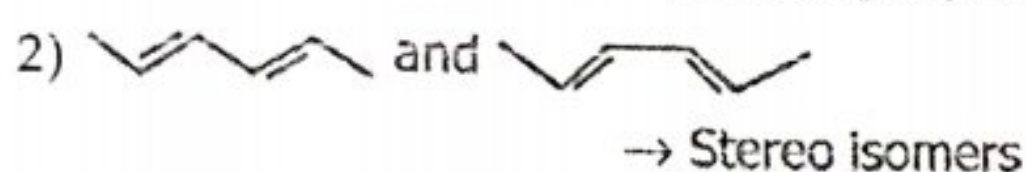
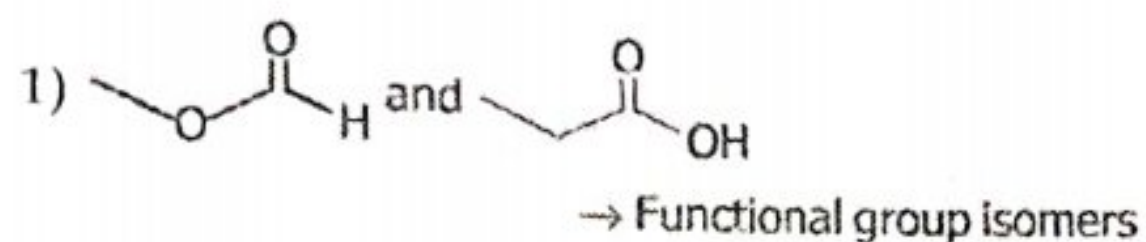
Which of the following relation is wrong for the compounds a and b

- i) $a > b$ (B.P.)
 - ii) $a = b$ (dipole moment)
 - iii) $a < b$ (solubility)
- 1) (i) and (iii)
 - 2) (i), (ii) and (iii)
 - 3) Only (ii)
 - 4) (i) and (ii)

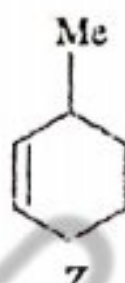
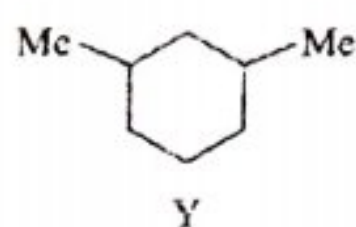
86. How many chiral centre are present in a molecule of 2, 4, 6-trimethyl heptane

- 1) 0
2) 1
3) 2
4) 3

87. Which of the following pair of isomers is correctly matched :

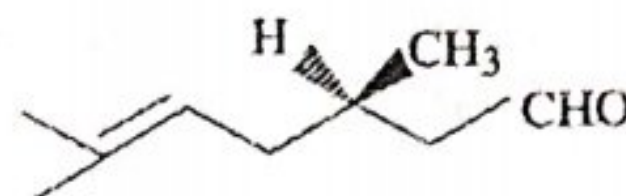
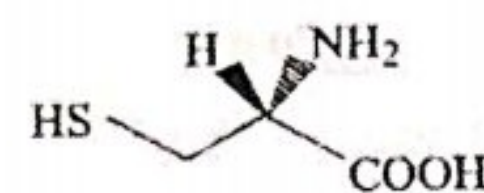
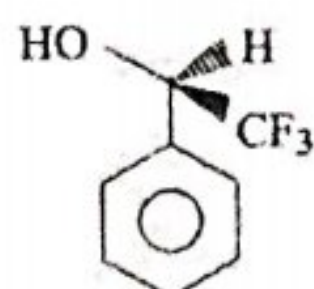


88. Number of chiral carbon atoms in the compound x, y and z respectively would be



- 1) 0, 2, 1
2) 1, 0, 1
3) 1, 2, 1
4) 1, 2, 0

89. The R/S configurations of these compounds are respectively



- 1) R, R, R
2) R, S, R
3) R, S, R
4) S, S, S

90. How many stereoisomers the following molecule are possible?



- 1) two optical isomers
2) two geometrical isomers
3) two optical and two geometrical isomers
4) none

SECTION 'C' BIOLOGY

91. A fall in GFR can activate the JG cells to release _____, which can stimulate the glomerular blood flow and thereby the GFR back to normal
 1) Renin 2) Angiotensin-II
 3) Rennin 4) Erythropoietin
92. Substances like glucose, amino acids, Na⁺ etc. in the filtrate are reabsorbed by
 1) Active transport
 2) Passive transport
 3) Both active and passive transport
 4) Facilitated diffusion
93. Osmotic concentration of glomerular filtrate is the highest at the bottom of the U-shaped Henle's loop. It is about ____mOsmL⁻¹
 1) 300 2) 600
 3) 900 4) 1200
94. Reabsorption of water from distal parts of the tubules is facilitated by hormone _____
 1) Vasopressin 2) ADH
 3) Aldosterone 4) Both (1) & (2)
95. Choose the incorrect statement
 1) Micturition is carried out by a reflex
 2) ADH helps in increased elimination, of water making the urine hypotonic
 3) Protein-free fluid is filtered from blood plasma into the Bowman's capsule
 4) Glucose is actively reabsorbed in the PCT
96. The check on RAAS is maintained by
 1) ADH 2) Aldosterone
 3) ANF 4) Renin
97. Ammonia and urea are waste product obtained from metabolic breakdown of
 1) Lipid 2) Carbohydrate
 3) Protein 4) Sugar
98. What happen if loop of henle become straight instead of hair pin like structure
 1) No effect on concentration of urine
 2) Water reabsorption increase
 3) Urine become more concentrated
 4) Urine become diluted
99. Statement - I : Medullary nephron have longer loop of henle, so they promote reabsorption of water from ascending limb
 Statement-II : Juxtaglomerular apparatus (JGA) made of efferent arterioles and DCT.
 1) Statement I and Statement II both are correct
 2) Statement I is incorrect but Statement II is correct
 3) Statement I and Statement II both incorrect
 4) Statement I is correct but Statement II is incorrect
100. Kidney function regulated by Renin-angiotensin system (RAS) then what happen if RAS inhibited by any drug
 1) Blood pressure (B.P.) increases
 2) Water loss in urine decrease
 3) Water loss in urine increases hence B.P. decrease
 4) B. P. decrease but GFR increases

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101. Which of the following is not incorrect about atrial natriuretic factor (ANF)
- 1) They increase B. P.
 - 2) They stimulate renin production
 - 3) They inhibit Renin-angiotensin system
 - 4) They are vasoconstrictor
102. Read the statement about counter current mechanisms
- a) It is regulated by loop of henle and afferent arterioles
 - b) Loop of henle and vasa recta play significant role
 - c) They maintain concentration gradient of medullary interstitium
 - d) In absence of counter-current, urine become hypertonic
- Select option which included all incorrect statement
- 1) a and c
 - 2) b and c
 - 3) a and d
 - 4) a and b
103. What quantity of blood on average is filtered by kidneys per minute ?
- 1) 3500 ml
 - 2) 1200 ml
 - 3) 5000 ml
 - 4) 650 ml
104. Glomerular filtrate at the base of vasa recta is (osmolarity) :-
- 1) 300 mOsmol L⁻¹
 - 2) 600 mOsmol L⁻¹
 - 3) 200 mOsmol L⁻¹
 - 4) 1200 mOsmol L⁻¹
105. Malfunctioning of kidneys can lead to accumulation of urea in blood, a condition called:-
- 1) Uremia
 - 2) Glomerulonephritis
 - 3) Renal calculi
 - 4) Ketonuria
106. Descending limb of loop of henle is permeable for :-
- 1) NaCl
 - 2) Water
 - 3) Urea
 - 4) NH₃
107. Which of the followings are secreted in PCT ?
- 1) Na⁺ & H⁺
 - 2) NH₃ & H⁺
 - 3) Cl⁻ & K⁺
 - 4) Na⁺ & HCO₃⁻
108. Which of the following substance is not excreted out in sweat ?
- 1) NaCl
 - 2) Small amount of urea
 - 3) Lactic acid
 - 4) Urochrome
109. Excretory organ in Amphioxus is
- 1) Flame cell
 - 2) Nephridia
 - 3) Renette cell
 - 4) Pronephric kidney
110. The solutes responsible for maintaining the high osmolarity of medulla are :
- 1) Na, Glucose
 - 2) NaCl, HCO₃⁻, Urea
 - 3) NaCl, K⁺, NH₃
 - 4) NaCl and Urea
111. Find out incorrect statement :
- 1) 25 gm of urea is excreted per day
 - 2) Renal column of bertini are the extensions of medulla into cortex
 - 3) Filtrate in PCT is isotonic
 - 4) Renal column of bertini are extension of cortex into medulla

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112. Read the following statements A to D for counter current mechanism :-

- A) The loop of Henle and vasa recta play a significant role in this.
 B) This mechanism helps to maintain a concentration gradient in the medullary interstitium.
 C) This mechanism helps to concentrate the urine.
 D) This mechanism occurs in all the nephrons of the kidneys.

How many statements are correct ?

- 1) One 2) Two
 3) Three 4) All

113. The skeletal system consists of

- 1) Framework of bones. 2) Few cartilages
 3) Both 1) and 2) 4) None of these

114. The skull is composed of

- 1) Cranial bones (8) 2) Facial bones (14)
 3) Both 1) and 2) 4) None of these

115. The hard and protective outer covering of brain is known as

- 1) Cranium 2) Condyle
 3) Meninges 4) All of these

116. The pectoral and pelvic girdles and the bones of limb form

- 1) Axial skeleton
 2) Appendicular skeleton
 3) Visceral skeleton
 4) Outer skeleton

117. The only movable bone of skull is

- 1) Mandible 2) Femure
 3) Sphenoid 4) Zygomatic

118. Amoeboid movement in macrophages and WBCs are effected by

- 1) Cilia 2) Flagella
 3) Pseudocoel 4) Pseudopodia

119. Match the column-I and II, and choose the correct combination from the options given.

	Column-I		Column-II
a.	Amoeboid movement	1	Trachea
b.	Ciliary movement	2	Spermatozoa
c.	Flagellar movement	3	Tongue
d.	Muscular movement	4	Neutrophils

- 1) (a) – (2), (b) – (3), (c) – (1), (d) – (4)
 2) (a) – (4), (b) – (2), (c) – (1), (d) – (3)
 3) (a) – (3), (b) – (1), (c) – (2), (d) – (4)
 4) (a) – (4), (b) – (1), (c) – (2), (d) – (3)

120. Passage of ova through the female reproductive tract is facilitated by the

- 1) Ciliary movement
 2) Amoeboid movement
 3) Muscular movement
 4) All of the above

121. Which of the following cytoskeletal elements involved in amoeboid movement ?

- 1) Microfilaments 2) Microtubules
 3) Myofibrils 4) More than 1 correct

122. Which of the following movement present in humans is not considered as a main type of movement in humans

- 1) Flagellar 2) Tentacular
 3) Ciliary 4) Muscular

123. Which of the following is/are the examples of flagella used for locomotion
- Euglena
 - Paramoecium
 - Sperm
 - Sponges
- a, b, c
 - a, c, d,
 - a, c
 - c and d
124. Which expired product of cellular respiration is eliminated through the lungs?"
- Ammonia
 - Urea
 - Carbon dioxide
 - Lactic acid
125. Assertion (A) : The DCT plays an important role in acid-base balance.
Reason (R): It is the site of bicarbonate secretion into the filtrate.
- Both A and R are true, and R is the correct explanation of A
 - Both A and R are true, but R is not the correct explanation of A
 - A is true, R is false
 - A is false, R is true
126. Identify the incorrect statements regarding JGA
- It is found near afferent arteriole
 - Macula densa cells secrete aldosterone
 - JG cells are modified smooth muscles
 - It plays a role in regulating GFR
- B only
 - A and D
 - C and D
 - A and B
127. In case of reduced afferent arteriole pressure:
- JGA is activated
 - Renin secretion is increased
 - Angiotensin II levels rise
 - GFR is irreversibly lost
- A, B, and C
 - A and D
 - All
 - B and D

128. Given below are two statement
- Statement I : A hairpin shaped Henle's loop which has a descending and an ascending limb
- Statement II : A large vessel of network runs "parallel to the Henle's loop forming a 'U'shaped "vasa recta.
- In the light of the above statements, choose the most appropriate answer from the options given below
- Both statement I and statement II are correct
 - Both statement I and statement II are incorrect
 - Statement I is correct but statement II is incorrect
 - Statement I is incorrect but statement II is correct
129. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)
- Assertion (A) : Urine formation involves 3 main "process"
- Reason (R) : It takes place in different parts of the nephron"
- In the light of the above statements, choose the most appropriate answer from the options given below
- Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - (A) is correct but (R) is not correct
 - (A) is not correct but (R) is correct
130. Given below are two statement
- Statement I : Glomerulus is a tuft of capillaries formed by the efferent arteriole
- Statement II : Glomerulus along with Bowman's capsule is called the malpighian body.

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In the light of the above statements, choose the most appropriate answer from the options given below

- 1) Both statement I and statement II are correct
- 2) Both statement I and statement II are incorrect
- 3) Statement I is correct but statement II is incorrect
- 4) Statement I is incorrect but statement II is correct

131. Given below are two statement

Statement I : Being soluble, ammonia is generally excreted in the form of NH_4^+

Statement II : Kidneys play a very important role in the excretion of ammonium ion.

In the light of the above statements, choose the most appropriate answer from the options given below

- 1) Both statement I and statement II are correct
- 2) Both statement I and statement II are incorrect
- 3) Statement I is correct but statement II is incorrect
- 4) Statement I is incorrect but statement II is correct

132. Find out incorrect match

- 1) ANF - Vasodilation
- 2) ADH - Vasodilation
- 3) Angiotension-II - Vasoconstriction
- 4) Vasopressin - Vasoconstriction

133. Among these which is not a function of kidney

- 1) Regulation of B.P
- 2) Removal of urea
- 3) Regulation of acidity of fluids

4) Secretion of antibiotics

134. Nitrogenous waste is not eliminated by

- 1) Skin
- 2) Kidney
- 3) Saliva
- 4) None

135. How many of below mentioned substances are secreted by liver? [Bilirubin, biliverdin, cholesterol, degraded steroid hormones, vitamins.

- 1) 3
- 2) 5
- 3) 4
- 4) 2

SECTION 'D' BIOLOGY

136. Chemiosmotic hypothesis :

- 1) Was given by Peter Mitchel
- 2) Is applicable for both mitochondrial and chloroplastic ATP synthesis
- 3) In chloroplast deals with ATP synthesis which is linked to the development of a proton gradient across the thylakoid membrane
- 4) Is related to all of the above

137. Urine release by

- 1) Relaxation of urethral sphincter and smooth muscles of bladder
- 2) Contraction of urethral sphincter and relaxation of smooth muscles of bladder
- 3) Contraction of both urethral sphincter and smooth muscles of bladder
- 4) Contraction of smooth muscles of bladder and relaxation of urethral sphincter

138. Urea is exchanged with collecting duct from which part of Henle's loop.

- 1) Thick part of descending limb
- 2) Thin part of descending limb
- 3) Thin part of ascending limb
- 4) Thick part of ascending limb

139. How many ATPs are formed per glucose molecule in Krebs' cycle?

- 1) 2 2) 24 3) 6 4) 28

140. Match the columns.

Column-I		Column-II	
A	Glucose → Glucose-6-phosphate	P	Phosphorylation by PFK
B	Fructose-6-phosphate → Fructose, 1,6-bisphosphate	Q	Hexokinase
C	Glyceraldehyde-3-phosphate → 1,3-bisphosphoglyceric acid	R	Formation of ATP
D	BPGA → 3-Phosphoglyceric acid	S	Formation of NADH + H ⁺

- 1) A-P, B-R, C-Q, D-S 2) A-S, B-Q, C-R, D-P
3) A-P, B-Q, C-R, D-S 4) A-Q, B-P, C-S, D-R

141. Kreb's cycle is

- 1) Oxidation of acetyl CoA to CO₂ and H₂O involving electron transport
- 2) Complete oxidation of acetyl CoA without electron transport
- 3) Oxidation of glucose to alcohol and water
- 4) Complete reduction of acetyl CoA to carbon dioxide and water

142. A value of RQ less than 1 means

- 1) Carbohydrates are used as respiratory material
- 2) Organic acids are used as respiratory substances
- 3) Oxidation of respiratory substrate consumes more O₂ than CO₂ evolved
- 4) Oxidation of respiratory substrate consumes less O₂ than CO₂ released

143. Which of the following statements of incorrect?

- 1) Oxidation-reduction reactions produce proton gradient in respiration
- 2) During aerobic respiration, role of oxygen is limited to the terminal state
- 3) In ETC (Electron Transport Chain), one molecule of NADH + H⁺ gives rise to 2 ATP molecules, and one FADH₂ gives rise to 3 ATP molecules
- 4) ATP is synthesized through complex V.

144. Assertion A : ATP is used at two steps in glycolysis.

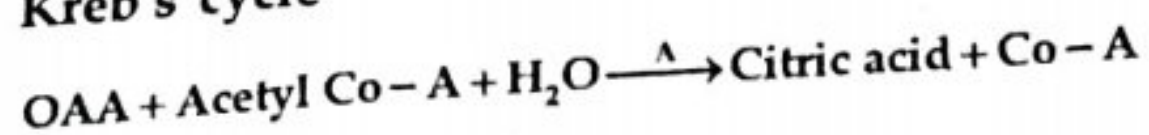
Reason R : First ATP is used in converting glucose into glucose-6-phosphate and second ATP is used in conversion of fructose-6-phosphate into fructose-1, 6-diphosphate.

- 1) A is false but R is true.
- 2) Both A and R are true and R is the correct explanation of A.
- 3) Both A and R are true but R is NOT the correct explanation of A.
- 4) A is true but R is false.

145. Where NADH + H⁺ is formed from NAD⁺ :

- 1) When fructose 6-phosphate converted to fructose 1, 6-disphosphate
- 2) When 3-phosphoglyceraldehyde (PGAL) is converted to 1, 3-phosphoglycerate (BPGA)
- 3) When 3-phosphoglyceraldehyde (PGAL) is converted to 1, 3-bisphosphoglycerate (BPGA)
- 4) When 3-phosphoglycerate (PGA) is converted to 3-phosphoglycerate (PGA)

146. Identify enzyme A in the given reaction of Kreb's cycle



- 1) Oxaloacetate synthetase
- 2) Citrate synthetase
- 3) Aconitase
- 4) Dehydrogenase

147. In glycolysis, fructose 1, 6 - bisphosphate get split into which of the following?

- 1) PGAL and BPGA
- 2) Glyceraldehyde - 3 - phosphate and 3 - phosphoglyceric acid
- 3) Glyceraldehyde - 3 - phosphate and Dihydroxy acetone phosphate
- 4) None of the above

148. The respiratory balance sheet is calculated on some assumptions.

Which of the following assumption is correct?

- 1) The pathway is sequential, with series of glycolysis, ETS and TCA cycle in the same order for a given molecule.
- 2) NADH is transferred to chloroplast where oxidative phosphorylation occurs, leading to formation of 3 ATP
- 3) Only glucose is the substrate and none other substrate or intermediate enters or leaves the pathway
- 4) None of these

149. Statement 1 : Substrate level phosphorylation takes place in glycolysis.

Statement 2 : It leads to synthesis of NADH_2

- 1) Both Statement I and Statement II are incorrect
- 2) Statement I is correct but Statement II is incorrect
- 3) Statement I is incorrect but Statement II is correct
- 4) Both Statement I and Statement II are correct

150. _____ : Chloroplast ; _____ : Glycolysis _____ : Aerobic respiration.

- 1) Photosynthesis, cytoplasm, mitochondria
- 2) Mitochondria, cytoplasm, mitochondria
- 3) Mitochondria, chloroplast, cytoplasm
- 4) Mitochondria, cytoplasm, chloroplast

151. Choose the wrong statement for Krebs' cycle.

- 1) There are four oxidation reactions in Krebs' cycle.
- 2) During conversion of succinyl CoA to succinic acid, a molecule of GTP is formed.
- 3) Decarboxylation occur at two steps.
- 4) The cycle starts with condensation reaction where Acetyl Co-A combines with pyruvic acid to form citric acid.

152. The enzymes of the steps of Krebs cycle where molecules of $\text{NADH} + \text{H}^+$ are produced, are

- 1) Isocitrate dehydrogenase, succinate dehydrogenase and malic dehydrogenase
- 2) Succinate thiokinase, succinate dehydrogenase and aconitase
- 3) Isocitrate dehydrogenase, α -ketoglutaric dehydrogenase, malic dehydrogenase
- 4) Isocitrate dehydrogenase, α -ketoglutaric dehydrogenase and succinate dehydrogenase

153. Match the columns.

Column-I		Column-II	
A	Breakdown of food materials within the cell	P	$6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy}$
B	Oxidation of carbohydrates	Q	Cellular respiration
C	Proteins, fats and even organic acids	R	To release energy
D	Complete combustion of glucose produces	S	Used as respiratory substances in some plants

- 1) A-R, B-P, C-Q, D-S 2) A-P, B-R, C-Q, D-S
3) A-Q, B-R, C-S, D-P 4) A-S, B-Q, C-R, D-P

154. Match the columns.

Column-I		Column-II	
A	End product of fermentation in yeast cells	P	2 molecules
B	Net ATP produced in fermentation	Q	NADH + H ⁺
C	End product of fermentation in muscle cells during exercise	R	Ethanol and CO ₂
D	Reducing agent in respiration process	S	Lactic acid and energy

- 1) A-P, B-Q, C-R, D-S 2) A-P, B-R, C-S, D-Q
3) A-R, B-P, C-S, D-Q 4) A-Q, B-R, C-S, D-P

155. In Kreb's cycle, which of the following reactions are involved in the production of NADH₂

1. Oxidation of succinate
2. Oxidation of alpha-ketoglutaric acid
3. Oxidation of isocitrate
4. Oxidation of malate
5. Oxidation of pyruvic acid to acetyl Co-A
6. Conversion of succinyl coenzyme-A to succinic acid

- 1) 1, 2, 6 2) 3, 2, 4
3) 1, 4, 3, 2, 6 4) 1, 2, 4

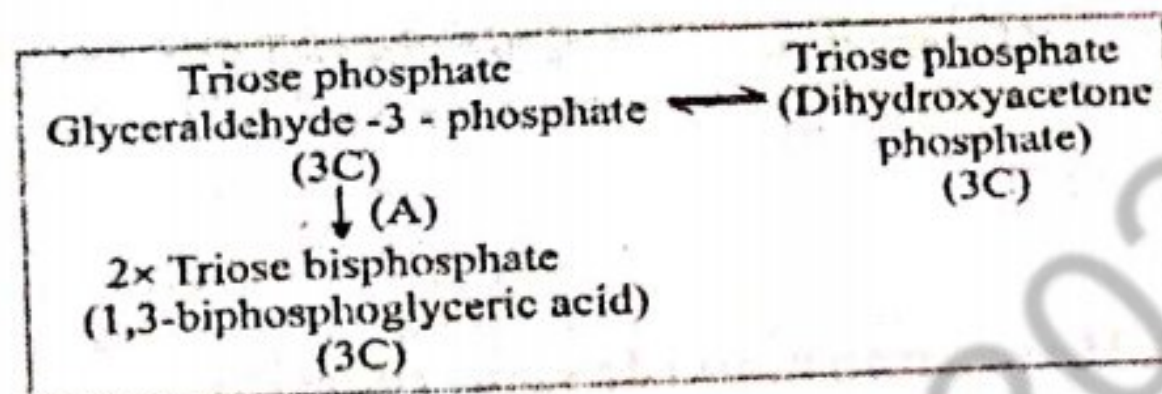
156. Assertion : In electron transport system, the electrons are passed on to oxygen, resulting in the formation of H₂O.

Reason : Oxygen is the ultimate acceptor of electrons.

- 1) Both assertion and reason are true and reason is correct explanation of assertion.

- 2) Both assertion and reason are true but reason is not the correct explanation of assertion.
3) Assertion is true but reason is false.
4) Both assertion and reason are false.

157. In the given diagram, label A represents -



- 1) ATP → ADP
2) ADP → ATP
3) NADH + H⁺ → NAD⁺
4) NAD⁺ → NADH + H⁺

158. Cytochrome c is a small protein attached to the [Old NCERT Pg. 233]

- 1) Outer surface of the inner membrane
- 2) Inner surface of the outer membrane
- 3) Inner surface of the inner membrane
- 4) Outer surface of the outer membrane

159. Match the columns I & II and choose the correct option.

Column - I		Column - II	
a.	Complex I	i.	Succinate dehydrogenase
b.	Complex II	ii.	NADH dehydrogenase
c.	Complex III	iii.	Cytochrome c oxidase
d.	Complex IV	iv.	Cytochrome bc ₁ complex

- 1) a-ii, b-i, c-iii, d-iv 2) a-iii, b-ii, c-iv, d-i
3) a-ii, b-i, c-iv, d-iii 4) a-i, b-ii, c-iii, d-iv

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QUESTION BOOKLET VERSION : 11

160. O_2 in areobic respiration acts as

- 1) Terminal hydrogen acceptor
- 2) Terminal electron acceptor
- 3) 1 and 2
- 4) Terminal water acceptor

161. Number of controlled steps required in glycolysis are :

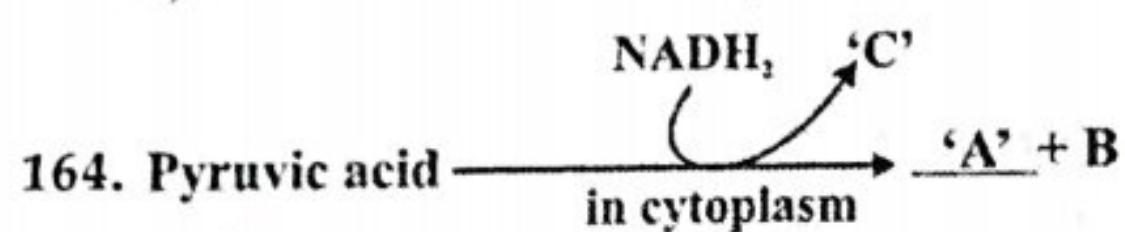
- 1) 1
- 2) 2
- 3) 10
- 4) 9

162. How many oxidation reaction occur in glycolysis?

- 1) 4
- 2) 2
- 3) 8
- 4) 6

163. Release of how many CO_2 molecules is common to form total 36 ATP with the help of mitochondrial inner membrane-

- 1) 12
- 2) 6
- 3) 3
- 4) 18



Choose the correct combination -

- 1) A = Lactic acid, B = CO_2 , C = NAD^+
- 2) A = Ethanol, B = H_2O , C = NAD^+
- 3) A = Ethanol, B = CO_2 , C = NAD^+
- 4) A = AcetylCoA, B = CO_2 , C = NAD^+

165. Fill in the blanks:

1. All living organisms need ...a... for canying out daily life activities, be it absorption, transport, movement, reproduction or even breathing.
2. All the energy required for life processes is obtained by ...b... some macromolecules that we call food.

3. Animals are heterotrophic, i.e they obtain food from plant directly (...c...) or indirectly (...d...).

4. ...e... like fungi are dependent on dead and decaying matter

1) a-food, b-oxidation, c-carnivores, d- herbivores, e-parasites

2) a-energy, b-reduction, c-herbivores, d- carnivores, e-saprophytes

3) a-energy, b-oxidation, c-herbivores, d- carnivores, e-saprophytes

4) a-oxygen, b-reduction, c-carnivores, d- herbivores, e-saprophytes

166. The crucial events in aerobic respiration are

1. The complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2

2. the passing on of the electrons removal as part of the hydrogen atoms to molecular O_2 with simultaneous synthesis of ATP

1) First process is ETS and takes place in matrix of the mitochondria while second process is TCA and takes place in on the inner membrane of mitochondria

2) First process is TCA and takes place in the matrix of mitochondria while the second process is ETS and is located on the inner membrane of mitochondria

3) First process is ETS and takes place on the inner membrane of mitochondria while second process is TCA and takes place in the matrix of the mitochondria

4) First process is TCA and takes place on the inner membrane of mitochondria while second process is ETS and takes place in the matrix of mitochondria

Space For Rough Work

$RCC^{**} RCC^{**} RCC^{**} RCC^{**} RCC^{**} RCC^{**} RCC^{**} RCC^{**} RCC^{**} RCC^{**}$

- 1) Krebs's cycle
- 2) Tricarboxylic acid cycle along with link reaction
- 3) Electron transport system
- 4) Both (1) and (3)

170. The energy utilised for the production of proton gradient required for phosphorylation in respiration is

- 1) Light energy
- 2) Energy of oxidation - reduction
- 3) photochemical energy
- 4) All of the above

171. Match the columns.

Column - I		Column - II	
a.	4C	i.	Acetyl CoA
b.	2C	ii.	Pyruvate
c.	5C	iii.	Citric acid
d.	3C	iv.	α -ketoglutaric acid
		v.	Malic acid

- 1) a-ii, b-v, c-iii, d-i 2) a-iii, b-i, c-iv, d-ii
3) a-v, b-i, c-iv, d-ii 4) a-v, b-iii, c-i, d-ii

172. Match the columns:

Column-I	Column-II
A. Joseph Priestley	(i) Green plant parts synthesise sugar
B. Jan Ingenhousz	(ii) Photosynthesis is light dependent reaction
C. Julius von Sachs	(iii) Discovered oxygen
D. Cornelius van Niel	(iv) Plants purify fouled air

- 1) A=(i), B=(iv), C=(iii), D=(ii)
- 2) A=(iv), B=(ii), C=(iii), D=(i)
- 3) A=(ii), B=(i), C=(iii), D=(iv)
- 4) A=(iii), B=(iv), C=(i), D=(ii)

- 1) Fermentation takes place under anaerobic condition in many prokaryotes & unicellular eukaryotes
- 2) A naturally fermented beverage can have 12% alcohol content
- 3) Krebs's cycle is also called tricarboxylic acid cycle
- 4) Total 2 molecules of $\text{NADPH} + \text{H}^+$ are used during glycolysis
- 5) ATP is produced in 3 steps in glycolysis

1) 2	2) 1
3) 3	4) 4

169. The complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2 . This statement refers to

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QUESTION BOOKLET VERSION : 11

173. Read the following statements :

- (i) In Engelmann's experiment the aerobic bacteria accumulated mainly in the region of green light of the split spectrum
- (ii) Action spectrum of photosynthesis corresponds closely to absorption spectra of chlorophyll-a showing that chlorophyll-a is the chief photosynthetic pigment
- (iii) Some organisms do not release O_2 during photosynthesis
- (iv) There occurs a clear division of labour w.r.t. light reaction and dark reaction within the chloroplast
- (v) Purple and green sulphur bacteria can liberate O_2 during photosynthesis

Which of the above statements are correct ?

- 1) (i), (ii) and (iii) 2) (i) and (v) only
- 3) (ii), (iii) and (iv) 4) (iii), (iv) and (v)

174. Cornelius van Neil :

- 1) Based on his studies of purple and green bacteria demonstrated that hydrogen from a suitable oxidisable compound oxidises CO_2 into carbohydrates
- 2) Demonstrated that photosynthesis is essentially a light independent reaction
- 3) Experimented only on green plants but not bacteria
- 4) Inferred that oxygen evolved by green plants come from H_2O

175. Which of the following experiment proves that CO_2 is essential for photosynthesis ?

- 1) Moll's half leaf experiment
- 2) Bell jar experiment
- 3) Thistle funnel experiment
- 4) Both (2) and (3)

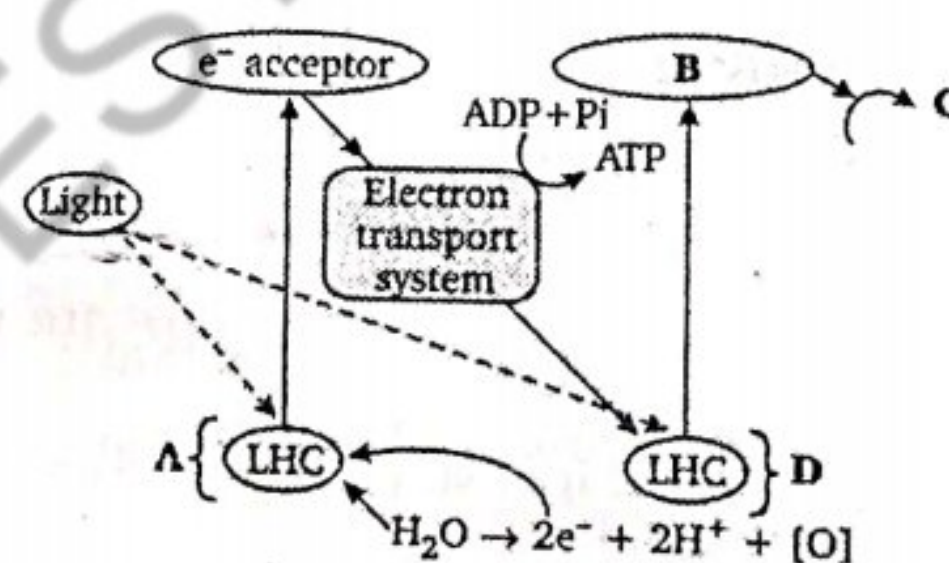
176. Read the following statements :

- (i) PS-I and PS-II were numbered I and II according to the sequence in which they function during light reaction
- (ii) The LHC are made of hundreds of pigment molecules bound to the proteins
- (iii) Many chlorophyll-a molecules form one reaction centre
- (iv) The reaction centre is different in PS-I and PS-II but with the same absorption peak of wavelength of light
- (v) In PS-II, the reaction centre chlorophyll-a gives out 680 nm wavelength of red light causing electrons to become excited and jump into an orbit farther from the atomic nucleus

Which of the above statement(s) are correct ?

- 1) (i), (ii) and (iii) 2) (ii) only
- 3) (v) only 4) (iii), (iv) and (v)

177. Observe the diagram given below and identify A, B, C and D.



A	B	C	D
1) PS-I	e^- acceptor	NANDH+ H^+	PS-II
2) PS-II	e^- acceptor	NANDP $^+$	PS-I
3) PS-II	e^- acceptor	NANDH+ H^+	PS-I
4) PS-I	e^- acceptor	FADH $_2$	PS-II

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QUESTION BOOKLET VERSION : 11

178. The water splitting complex is :

- 1) Associated with the PS-II
- 2) Located on the inner side of the thylakoid membrane
- 3) May remain associated with PS-I
- 4) More than one option is correct

179. Read the following statements :

- (i) During non-cyclic electron flow (Z-scheme) both ATP and $\text{NADH} + \text{H}^+$ are synthesised in higher plants
- (ii) During non-cyclic electron transport only PS-I is functional
- (iii) Stroma lamella (fret) membranes lack PS-II as well as $\text{NADP}^+ - \text{reductase}$
- (iv) During cyclic electron transport chain only ATP is synthesised but not $\text{NADPH} + \text{H}^+$
- (v) Cyclic photophosphorylation occurs when only light of wavelengths beyond 680 nm are available for excitation

Which of the above statements are correct ?

- 1) (iii), (iv) and (v) 2) (i), (ii) and (iii)
- 3) (ii) and (iii) 4) (i) and (ii)

180. which of the following is correct ?

- 1) PS-I is involved in non-cyclic photophosphorylation but not cyclic photophosphorylation
- 2) PS-II is involved in both non-cyclic as well as cyclic photophosphorylation
- 3) PS-II is involved only in non-cyclic oxidative phosphorylation
- 4) PS-I is involved in both non-cyclic as well as cyclic photophosphorylation

Space For Rough Work

RCC

Physics		Chemistry			Biology				
01. 1	19. 1	37. 2	55. 1	73. 2	91. 1	109.1	127.1	145.3	163.2
02. 2	20. 4	38. 3	56. 3	74. 4	92. 1	110.4	128.1	146.2	164.3
03. 4	21. 4	39. 3	57. 4	75. 1	93. 4	111.2	129.2	147.3	165.3
04. 4	22. 1	40. 3	58. 4	76. 1	94. 4	112.3	130.4	148.3	166.2
05. 4	23. 3	41. 4	59. 3	77. 1	95. 2	113.3	131.3	149.2	167.2
06. 1	24. 4	42. 4	60. 3	78. 3	96. 3	114.3	132.2	150.1	168.1
07. 1	25. 1	43. 3	61. 3	79. 2	97. 3	115.1	133.4	151.4	169.2
08. 1	26. 3	44. 4	62. 3	80. 2	98. 4	116.2	134.4	152.3	170.2
09. 1	27. 1	45. 3	63. 3	81. 3	99. 3	117.1	135.2	153.3	171.3
10. 1	28. 3	46. 3	64. 4	82. 1	100.3	118.4	136.4	154.3	172.4
11. 1	29. 2	47. 4	65. 3	83. 4	101.3	119.4	137.4	155.2	173.3
12. 4	30. 2	48. 3	66. 2	84. 2	102.3	120.1	138.3	156.1	174.4
13. 4	31. 2	49. 2	67. 3	85. 4	103.2	121.1	139.2	157.4	175.1
14. 2	32. 2	50. 1	68. 2	86. 1	104.4	122.1	140.4	158.1	176.2
15. 2	33. 1	51. 4	69. 1	87. 4	105.1	123.3	141.2	159.3	177.3
16. 3	34. 2	52. 4	70. 4	88. 1	106.2	124.3	142.3	160.3	178.4
17. 1	35. 3	53. 1	71. 2	89. 1	107.2	125.3	143.3	161.3	179.1
18. 4	36. 1	54. 2	72. 3	90. 1	108.4	126.1	144.2	162.2	180.4